

POCKETSMART AI

Requirement Analysis Phase

Project Name	PocketSmart AI
Phase	Requirement Analysis
Project Type	Artificial Intelligence / Smart Assistant

1. Introduction

Requirement Analysis is the phase in which the requirements of PocketSmart AI are identified and documented. It defines what the system should do, who will use it, and the technical and functional requirements needed to develop the application successfully.

2. Purpose of the System

The purpose of PocketSmart AI is to provide an easy-to-use AI-powered assistant that can understand user requests, provide useful responses, and support everyday productivity through smart assistance.

3. Target Users

- Students who need help organizing daily activities.
- Working professionals who need quick assistance.
- General users who want an AI-based personal assistant.

4. Functional Requirements

- The system shall provide an interface for entering user requests.
- The system shall process natural-language user input.
- The system shall generate an appropriate AI-based response.
- The system shall support task and reminder-related requests.
- The system shall display responses clearly to the user.
- The system shall handle invalid or unclear input with a suitable message.
- The system should allow future features to be integrated easily.

5. Non-Functional Requirements

- Usability – The interface should be simple and easy to understand.
- Performance – Responses should be generated within a reasonable time.
- Reliability – The application should work consistently for valid requests.
- Security – User information and API credentials should be handled securely.

- Scalability – The system should support additional AI features in future.
- Maintainability – The code should be organized and easy to update.

6. Hardware Requirements

Component	Minimum Requirement
Processor	Dual-core processor or above
RAM	4 GB or above
Storage	At least 2 GB free space
Internet	Required for AI/API-based features
Display	Standard monitor or laptop display

7. Software Requirements

- Operating System: Windows / Linux / macOS
- Programming Language: Python
- AI Service/API: Gemini or another supported generative AI model
- Development Environment: Visual Studio Code or equivalent IDE
- Required Python libraries and dependencies as defined by the project implementation
- Web browser for accessing the application when deployed as a web-based system

8. Input and Output Requirements

Type	Requirement
Input	User questions, requests, tasks, reminders, and other natural-language prompts.
Processing	AI model analyzes the input and generates a relevant response.
Output	Clear AI-generated response or task-related information displayed to the user.

9. Constraints and Assumptions

- An active internet connection may be required for cloud-based AI services.
- AI responses depend on the capabilities and availability of the selected AI model.
- API keys or credentials must be configured securely.
- The first version may provide a limited set of assistant functions.

10. Expected Outcome

The requirement analysis provides a clear specification for developing PocketSmart AI. It establishes the required features, user needs, system requirements and technical expectations that will guide the Project Design and Development phases.

11. Conclusion

The Requirement Analysis phase converts the initial PocketSmart AI idea into a structured set of requirements. These requirements form the foundation for designing and implementing a useful, reliable and user-friendly AI assistant.